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HAND TOOL HOLDER

Abstract

A hand tool holder includes a holder body including a first face, a second face, a third face, and a fourth face. The holder body is provided with at least one first groove arranged on the second face and the third face. The at least one first groove includes a first recessed portion, a first connecting portion, and a first mounting portion which are connected to construct an L-shaped channel. The holder body is provided with at least one second mounting portion connected to the first connecting portion. The holder body is provided with a fifth face. The at least one second mounting portion has an opening connected to the fifth face. The first recessed portion has an opening connected to the fifth face.

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

[0001] The present invention relates to a holder (or rack) and, more particularly, to a hand tool holder.

Description of the Related Art

[0002] A conventional hand tool set combination was disclosed in the U.S. Pat. No. 8,387,492, and comprises a grip body, two support members, and multiple tool parts. The grip body includes two side covers. Each of the side covers has a storage space. Each of the side covers has an end provided with a pivot portion. Each of the side covers has a face provided with an elongate slot. The support members are received in the side covers. Each of the support members has a pivot shaft pivotally connected with the pivot portion of each of the side covers. Each of the tool parts has a pivot end pivotally connected with the pivot shaft of each of the support members. Each of the tool parts has a distal end formed with a tool end. An extension rod is formed between the pivot end and the tool end and extends through the elongate slot of each of the side covers.

[0003] In operation, by the quick release structure of the two side covers and the rotational pivoting of the tool parts, the tool end of each of the tool parts extends outward from the elongate slot of each of the side covers to be operated by the user. Alternatively, each of the tool parts is received and stored in the storage space of each of the side covers.

[0004] However, such a conventional hand tool set combination has the following disadvantages.

[0005] 1. The pivot end of each of the tool parts is pivotally connected with the pivot shaft of each of the support members, so that the extension rod of each of the tool parts extends through the elongate slot of each of the side covers through 90 degrees or 180 degrees only. Thus, the assembly position of each of the tool parts is limited by each of the support members, and the movement of each of the tool parts is restricted, so that each of the tool parts can only be rotated through a determined angle. [0006] 2. The grip body is only pivotally connected with the tool parts and cannot be assembled with other tool parts, thereby limiting the versatility of the grip body. [0007] 3. The pivot end of each of the tool parts is pivotally connected with the pivot shaft of each of the support members, so that each of the tool parts cannot be detached from the grip body, thereby limiting the mobility of each of the tool parts. [0008] 4. Each of the tool parts is pivotally mounted on the pivot shaft of each of the support members, and each of the tool parts is rotated outward from the elongate slot of each of the side covers to be operated by the user, so that the axis of each of the tool parts is easily rotated and shifted during operation. Thus, each of the tool parts will be able to rotate back automatically, thereby causing inconvenience to the user during operation.

BRIEF SUMMARY OF THE INVENTION

[0009] In accordance with the present invention, there is provided a hand tool holder comprising a holder body. The holder body includes a first face, a second face, a third face, and a fourth face. The holder body is provided with at least one first groove arranged on the second face and the third face. The at least one first groove has an L-shaped profile. The at least one first groove includes a first recessed portion, a first connecting portion, and a first mounting portion. The first recessed portion, the first connecting portion, and the first mounting portion are connected and construct an L-shaped channel. The holder body is provided with at least one second mounting portion located beside the at least one first groove. The at least one second mounting portion and the first recessed portion extend in the same direction. The at least one second mounting portion is connected to the first connecting portion. The holder body is provided with a fifth face. The at least one second mounting portion has an opening connected to the fifth face. The first recessed portion has an opening connected to the fifth face.

[0010] According to the primary advantages of the present invention, the holder body is provided with the at least one first groove and the at least one second mounting portion for mounting the at least one first tool part so that the holder body is served as a handle to operate the at least one first

tool part. Thus, the holder body is used to store the at least one first tool part and also used to provide multiple operational modes to the at least one first tool part.

[0011] According to another advantage of the present invention, the at least one first tool part is removably mounted on the at least one first groove or the at least one second mounting portion, so that the at least one first tool part is mounted on different positions of the holder body to provide various operational modes. In addition, the at least one first tool part is removed from the holder body and operated independently.

[0012] Further benefits and advantages of the present invention will become apparent after a careful reading of the detailed description with appropriate reference to the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWING(S)

[0013] FIG. 1 is an exploded perspective view of a hand tool holder in accordance with the preferred embodiment of the present invention.

[0014] FIG. 2 is a perspective view of a holder body of the hand tool holder in accordance with the preferred embodiment of the present invention.

[0015] FIG. 3 is a top view of the holder body of the hand tool holder as shown in FIG. 2.

[0016] FIG. 4 is another perspective view of the holder body of the hand tool holder in accordance with the preferred embodiment of the present invention.

[0017] FIG. 5 is a bottom view of the holder body of the hand tool holder as shown in FIG. 2.

[0018] FIG. 6 is a perspective view of the hand tool holder in accordance with the preferred embodiment of the present invention.

[0019] FIG. 7 is a perspective view showing a first operational state of the hand tool holder in accordance with the preferred embodiment of the present invention.

[0020] FIG. 8 is a perspective view showing a second operational state of the hand tool holder in accordance with the preferred embodiment of the present invention.

[0021] FIG. 9 is a perspective view showing a third operational state of the hand tool holder in accordance with the preferred embodiment of the present invention.

[0022] FIG. 10 is a perspective view showing a fourth operational state of the hand tool holder in accordance with the preferred embodiment of the present invention.

[0023] FIG. 11 is an exploded perspective view of a hand tool holder in accordance with the second preferred embodiment of the present invention.

[0024] FIG. 12 is a perspective view of a holder body of the hand tool holder in accordance with the second preferred embodiment of the present invention.

[0025] FIG. 13 is a perspective view showing an operation state of the hand tool holder in accordance with the second preferred embodiment of the present invention.

[0026] FIG. 14 is an exploded perspective view of a hand tool holder in accordance with the third preferred embodiment of the present invention.

DETAILED DESCRIPTION OF THE INVENTION

[0027] Referring to the drawings and initially to FIGS. 1-6, a hand tool holder in accordance with the preferred embodiment of the present invention comprises a holder body 10. The holder body 10 includes a first face 101, a second face 102, a third face 103, and a fourth face 104. Each of the first face 101 and the third face 103 is longer than the second face 102 and the fourth face 104.

[0028] The holder body 10 is provided with at least one first groove 11 arranged on the second face 102 and the third face 103. The at least one first groove 11 has an L-shaped profile. Preferably, the holder body 10 is provided with multiple first grooves 11 having different sizes. Optimally, the holder body 10 is provided with three first grooves 11. The at least one first groove 11 includes a

first recessed portion **111**, a first connecting portion **112**, and a first mounting portion **113**. The first recessed portion **111**, the first connecting portion **112**, and the first mounting portion **113** are connected and construct an L-shaped channel. The first recessed portion **111** is open at the second face **102**. The first connecting portion **112** is open at the third face **103** and is located at a bent position of the at least one first groove **11**. The first mounting portion **113** is closed at the third face **103** and is perpendicular to the first recessed portion **111**. The first mounting portion **113** is a polygonal or hexagonal hole in cross-section.

[0029] The holder body **10** is provided with at least one second mounting portion **12** located beside the at least one first groove **11**. The at least one second mounting portion **12** and the first recessed portion **111** extend in the same direction. The at least one second mounting portion **12** is connected to the first connecting portion **112** and has a specification equal to that of the first mounting portion **113**. The at least one second mounting portion **12** is a polygonal or hexagonal hole penetrating the holder body **10** longitudinally. Preferably, the holder body **10** is provided with multiple second mounting portions **12** having different sizes. The second mounting portions **12** have a number equal to that of the first grooves **11**. Optimally, the holder body **10** is provided with three second mounting portions **12**. The holder body **10** is provided with a fifth face **13**. The at least one second mounting portion **12** has an opening connected to the fifth face **13**. The first recessed portion **111** has an opening connected to the fifth face **13**. The fifth face **13** has a height different from and lower than that of the first face **101**.

[0030] The holder body **10** is provided with a first space **14** having an opening connected to the first face **101**. The first space **14** is an elongate slot with one open side. The first space **14** is used to store hand tool parts with smaller specifications, such as sockets or the like.

[0031] The holder body **10** is provided with at least one second groove **15** arranged on the first face **101** and the fourth face **104**. The at least one second groove **15** has an L-shaped profile. Preferably, the holder body **10** is provided with multiple second grooves **15** having different sizes. Optimally, the holder body **10** is provided with five second grooves **15**. The at least one second groove **15** includes a second recessed portion **151**, a second connecting portion **152**, and a third mounting portion **153**. The second recessed portion **151**, the second connecting portion **152**, and the third mounting portion **153** are connected and construct an L-shaped channel. The second recessed portion **151** is open at the fourth face **104**. The second connecting portion **152** is open at the first face **101** and is located at a bent position of the at least one second groove **15**. The third mounting portion **153** is closed at the first face **101** and is perpendicular to the second recessed portion **151**. The third mounting portion **153** is a polygonal or hexagonal hole in cross-section.

[0032] The holder body **10** is provided with at least one fourth mounting portion **16** located beside the at least one second groove **15**. The at least one fourth mounting portion **16** and the second recessed portion **151** extend in the same direction. The at least one fourth mounting portion **16** is connected to the second connecting portion **152** and has a specification equal to that of the third mounting portion **153**. The at least one fourth mounting portion **16** is a polygonal or hexagonal hole penetrating the holder body **10** longitudinally. Preferably, the holder body **10** is provided with multiple fourth mounting portions **16** having different sizes. The fourth mounting portions **16** have a number equal to that of the second groove **15**. Optimally, the holder body **10** is provided with five fourth mounting portions **16**. The holder body **10** is provided with a sixth face **17**. The at least one fourth mounting portion **16** has an opening connected to the sixth face **17**. The second recessed portion **151** has an opening connected to the sixth face **17**. The sixth face **17** has a height different from and lower than that of the third face **103**.

[0033] The holder body **10** is provided with a second space **18** having an opening connected to the third face **103**.

[0034] The first mounting portion **113** is connected to the second space **18** and is a perforation penetrating the holder body **10**. The third mounting portion **153** is connected to the first space **14** and is a perforation penetrating the holder body **10**.

[0035] The hand tool holder further comprises at least one first tool part **20** including a hex wrench structure. The at least one first tool part **20** is received in the at least one first groove **11** or the at least one second mounting portion **12**. The at least one first tool part **20** has a first end **21** and a second end **22**. The first end **21** is received in the first recessed portion **111**. The second end **22** is received in the first connecting portion **112** and the first mounting portion **113**. The first end **21** partially protrudes from the first recessed portion **111** and partially protrudes from the fifth face **13**. Preferably, the hand tool holder has multiple first tool parts **20** having different specifications.

[0036] After the at least one first tool part **20** is mounted in the at least one first groove **11**, the second end **22** extends into the second space **18** and slightly closes or seals an opening of the second space **18**.

[0037] The hand tool holder further comprises at least one second tool part **30**. The at least one second tool part **30** is received in the at least one second groove **15** or the at least one fourth mounting portion **16**. The at least one second tool part **30** has an end partially protruding from the second recessed portion **151** and partially protruding from the sixth face **17**. Preferably, the hand tool holder has multiple second tool parts **30** having different specifications.

[0038] As shown in FIG. **6**, the at least one first tool part **20** is mounted in the at least one first groove **11**, and the at least one second tool part **30** is mounted in the at least one second groove **15**. Thus, the holder body **10** is used to hold the at least one first tool part **20** and the at least one second tool part **30**, to facilitate the user storing and carrying the at least one first tool part **20** and the at least one second tool part **30**.

[0039] Referring to FIG. **7** with reference to FIGS. **1-6**, the first end **21** partially protrudes from the first recessed portion **111**. Thus, the first end **21** is pulled out by the user's finger from the fifth face **13**, so that the second end **22** is pulled outward from the first mounting portion **113**. In such a manner, the holder body **10** is held and driven by the user to rotate the at least one first tool part **20**. Thus, the at least one first tool part **20** has a first operational mode.

[0040] Referring to FIG. **8** with reference to FIGS. **1-6**, the at least one first tool part **20** is pulled outward and detached from the at least one first groove **11**. Then, the at least one first tool part **20** is turned, with the first end **21** being inserted into the first mounting portion **113**, and with the second end **22** protruding from the at least one first groove **11**. In such a manner, the holder body **10** is served as a handle to rotate the second end **22**. Thus, the at least one first tool part **20** has a second operational mode.

[0041] Referring to FIG. **9** with reference to FIGS. **1-6**, the at least one first tool part **20** is pulled outward and detached from the at least one first groove **11**. Then, the first end **21** is inserted into the at least one second mounting portion **12**, and the second end **22** protrudes from the at least one second mounting portion **12**. Thus, the at least one first tool part **20** has a third operational mode.

[0042] Referring to FIG. **10** with reference to FIGS. **1-6**, the at least one first tool part **20** is pulled outward and detached from the at least one first groove **11**. Then, the at least one first tool part **20** is placed into the second space **18**, with the first end **21** being inserted into the first mounting portion **113**, and with the second end **22** protruding from the second space **18**. Thus, the at least one first tool part **20** has a fourth operational mode.

[0043] In conclusion, referring to FIGS. **7-10** with reference to FIGS. **1-6**, the first end **21** or the second end **22** is inserted into and mounted in the first mounting portion **113** or the at least one second mounting portion **12**. Each of the first mounting portion **113** and the at least one second mounting portion **12** is a perforation penetrating the holder body **10**, so that the first end **21** or the second end **22** is inserted into the first mounting portion **113** or the at least one second mounting portion **12** in two different directions. Thus, the at least one first tool part **20** is mounted on the holder body **10** to perform various operational modes.

[0044] Similarly, the at least one second tool part **30** is mounted on the holder body **10** to perform various operational modes similar to that of the at least one first tool part **20**.

[0045] Referring to FIGS. **11** and **12**, the holder body **10** is provided with a fifth mounting portion

19 formed in the first space **14**. The fifth mounting portion **19** is a square block. The hand tool holder further comprises a third tool part **40** received in the first space **14**. After the third tool part **40** is removed from the first space **14**, the third tool part **40** is mounted on the fifth mounting portion **19**. The holder body **10** is provided with two first slideways **191** arranged on two sides of the second space **18**. The hand tool holder further comprises a fourth tool part **50** provided with two second slideways **511** slidably mounted on the two first slideways **191** so that the fourth tool part **50** is received in the second space **18** or removed from the second space **18**.

[0046] Referring to FIG. **13** with reference to FIG. **11**, the first tool parts **20** (or the second tool parts **30**) are removed from the holder body **10**. Then, a first one of the first tool parts **20** (or the second tool parts **30**) is inserted into one mounting recess of the third tool part **40**, and a second one of the first tool parts **20** (or the second tool parts **30**) is inserted into another mounting recess of the third tool part **40**. Thus, the third tool part **40** is driven by the first one of the first tool parts **20** (or the second tool parts **30**) to rotate the second one of the first tool parts **20** (or the second tool parts **30**).

[0047] Referring to FIG. **14** with reference to FIGS. **1-6**, the at least one second groove **15**, the at least one fourth mounting portion **16**, and the sixth face **17** of the holder body **10** are undefined. In addition, the at least one second tool part **30** is undefined.

[0048] In another operational mode of the present invention, the at least one first tool part **20** and the at least one second tool part **30** are removed from the holder body **10**. Thus, the at least one first tool part **20** and the at least one second tool part **30** are operated independently.

[0049] In another preferred embodiment of the present invention, the at least one first tool part **20** is an L-shaped shank received in the at least one first groove **11**.

[0050] Accordingly, the hand tool holder of the present invention has the following advantages.

[0051] 1. The holder body **10** is provided with the at least one first groove **11** and the at least one second mounting portion **12** for mounting the at least one first tool part **20** so that the holder body **10** is served as a handle to operate the at least one first tool part **20**. Thus, the holder body **10** is used to store the at least one first tool part **20** and also used to provide multiple operational modes to the at least one first tool part **20**. [0052] 2. The at least one first tool part **20** is removably mounted on the at least one first groove **11** or the at least one second mounting portion **12**, so that the at least one first tool part **20** is mounted on different positions of the holder body **10** to provide various operational modes. In addition, the at least one first tool part **20** is removed from the holder body **10** and operated independently. [0053] 3. The first end **21** or the second end **22** is inserted into and mounted in the first mounting portion **113** or the at least one second mounting portion **12**. Each of the first mounting portion **113** and the at least one second mounting portion **12** is a perforation penetrating the holder body **10**, so that the first end **21** or the second end **22** is inserted into the first mounting portion **113** or the at least one second mounting portion **12** in two different directions. Thus, the at least one first tool part **20** is mounted on the holder body **10** to perform various operational modes. [0054] 4. The holder body **10** is provided with the at least one second groove **15** and the at least one fourth mounting portion **16** for mounting the at least one second tool part **30** so that the holder body **10** is served as a handle to operate the at least one second tool part **30**. Thus, the holder body **10** is used to store the at least one second tool part **30** and also used to provide multiple operational modes to the at least one second tool part **30**. [0055] 5. The at least one second tool part **30** is removably mounted on the at least one second groove **15** or the at least one fourth mounting portion **16**, so that the at least one second tool part **30** is mounted on different positions of the holder body **10** to provide various operational modes. In addition, the at least one second tool part **30** is removed from the holder body **10** and operated independently. [0056] 6. The holder body **10** is used to store the at least one first tool part **20** and the at least one second tool part **30**. The at least one first tool part **20** and the at least one second tool part **30** are respectively mounted on different positions of the holder body **10** to provide various operational modes. In addition, the at least one first tool part **20** and the at least one second tool part **30** are respectively removed from

the holder body **10** and operated independently. [0057] 7. The at least one first tool part **20** and the at least one second tool part **30** are respectively mounted on the first mounting portion **113**, the at least one second mounting portion **12**, the third mounting portion **153**, and the at least one fourth mounting portion **16** by a hexagonal engagement, so that the at least one first tool part **20** and the at least one second tool part **30** are operated steadily without slipping. [0058] 8. The holder body **10** is provided with the fifth mounting portion **19** and the two first slideways **191**. The third tool part **40** is mounted on the fifth mounting portion **19** so that the third tool part **40** is received and stored in the first space **14**. The fourth tool part **50** is provided with the two second slideways **511** slidably mounted on the two first slideways **191** so that the fourth tool part **50** is received and stored in the second space **18**. [0059] 9. Referring to FIG. **13**, the first tool parts **20** (or the second tool parts **30**) are removed from the holder body **10** completely. Then, a first one of the first tool parts **20** (or the second tool parts **30**) is assembled with the third tool part **40**, and a second one of the first tool parts **20** (or the second tool parts **30**) is assembled with the third tool part **40**. Thus, the third tool part **40** is driven by the first one of the first tool parts **20** (or the second tool parts **30**) to rotate the second one of the first tool parts **20** (or the second tool parts **30**).

[0060] Although the invention has been explained in relation to its preferred embodiment(s) as mentioned above, it is to be understood that many other possible modifications and variations can be made without departing from the scope of the present invention. It is, therefore, contemplated that the appended claim or claims will cover such modifications and variations that fall within the scope of the invention.

Claims

1. A hand tool holder comprising: a holder body; wherein: the holder body includes a first face, a second face, a third face, and a fourth face; each of the first face and the third face is longer than the second face and the fourth face; the holder body is provided with at least one first groove arranged on the second face and the third face; the at least one first groove has an L-shaped profile; the at least one first groove includes a first recessed portion, a first connecting portion, and a first mounting portion; the first recessed portion, the first connecting portion, and the first mounting portion are connected and construct an L-shaped channel; the first recessed portion is open at the second face; the first connecting portion is open at the third face and is located at a bent position of the at least one first groove; the first mounting portion is closed at the third face and is perpendicular to the first recessed portion; the first mounting portion is a polygonal or hexagonal hole in cross-section; the holder body is provided with at least one second mounting portion located beside the at least one first groove; the at least one second mounting portion and the first recessed portion extend in the same direction; the at least one second mounting portion is connected to the first connecting portion and has a specification equal to that of the first mounting portion; the at least one second mounting portion is a polygonal or hexagonal hole penetrating the holder body longitudinally; the holder body is provided with a fifth face; the at least one second mounting portion has an opening connected to the fifth face; the first recessed portion has an opening connected to the fifth face; the fifth face has a height different from and lower than that of the first face; the holder body is provided with a first space having an opening connected to the first face; the first space is an elongate slot with one open side; the holder body is provided with at least one second groove arranged on the first face and the fourth face; the at least one second groove has an L-shaped profile; the at least one second groove includes a second recessed portion, a second connecting portion, and a third mounting portion; the second recessed portion, the second connecting portion, and the third mounting portion are connected and construct an L-shaped channel; the second recessed portion is open at the fourth face; the second connecting portion is open at the first face and is located at a bent position of the at least one second groove; the third mounting portion is closed at the first face and is perpendicular to the second recessed portion; the

third mounting portion is a polygonal or hexagonal hole in cross-section; the holder body is provided with at least one fourth mounting portion located beside the at least one second groove; the at least one fourth mounting portion and the second recessed portion extend in the same direction; the at least one fourth mounting portion is connected to the second connecting portion and has a specification equal to that of the third mounting portion; the at least one fourth mounting portion is a polygonal or hexagonal hole penetrating the holder body longitudinally; the holder body is provided with a sixth face; the at least one fourth mounting portion has an opening connected to the sixth face; the second recessed portion has an opening connected to the sixth face; the sixth face has a height different from and lower than that of the third face; the holder body is provided with a second space having an opening connected to the third face; the first mounting portion is connected to the second space and is a perforation penetrating the holder body; and the third mounting portion is connected to the first space and is a perforation penetrating the holder body.

2. The hand tool holder as claimed in claim 1, wherein: the holder body is provided with multiple first grooves having different sizes; and the holder body is provided with three first grooves.

3. The hand tool holder as claimed in claim 1, wherein: the holder body is provided with multiple second mounting portions having different sizes; the second mounting portions have a number equal to that of the first grooves; and the holder body is provided with three second mounting portions.

4. The hand tool holder as claimed in claim 1, wherein: the holder body is provided with multiple second grooves having different sizes; the holder body is provided with five second grooves; the holder body is provided with multiple fourth mounting portions having different sizes; the fourth mounting portions have a number equal to that of the second groove; and the holder body is provided with five fourth mounting portions.

5. The hand tool holder as claimed in claim 1, wherein: the hand tool holder further comprises at least one first tool part including a hex wrench structure; the at least one first tool part is received in the at least one first groove or the at least one second mounting portion; the at least one first tool part has a first end and a second end; the first end is received in the first recessed portion; the second end is received in the first connecting portion and the first mounting portion; the first end partially protrudes from the first recessed portion and partially protrudes from the fifth face; and the hand tool holder has multiple first tool parts having different specifications.

6. The hand tool holder as claimed in claim 5, wherein after the at least one first tool part is mounted in the at least one first groove, the second end extends into the second space and slightly closes an opening of the second space.

7. The hand tool holder as claimed in claim 5, further comprising: at least one second tool part; wherein: the at least one second tool part is received in the at least one second groove or the at least one fourth mounting portion; the at least one second tool part has an end partially protruding from the second recessed portion and partially protruding from the sixth face; and the hand tool holder has multiple second tool parts having different specifications.

8. The hand tool holder as claimed in claim 1, wherein: the holder body is provided with a fifth mounting portion formed in the first space; the fifth mounting portion is a square block; the hand tool holder further comprises a third tool part received in the first space; after the third tool part is removed from the first space, the third tool part is mounted on the fifth mounting portion; the holder body is provided with two first slideways arranged on two sides of the second space; and the hand tool holder further comprises a fourth tool part provided with two second slideways slidably mounted on the two first slideways so that the fourth tool part is received in the second space or removed from the second space.
